

IF330 Pemrograman Web.

05 –PHP File Operation

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PROGRAM STUDI INFORMATIKA
FAKULTAS TEKNIK DAN INFORMATIKA
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REVIEW



- Object-Oriented Programming
- Classes & Objects
- Constructor & Destructor
- Encapsulation / Access Modifier
- Inheritance & Final keyword
- Constants
- Abstract Classes
- Interface
- Traits
- Static Properties & Methods
- Autoloading
- Namespaces

TODAY's OUTLINE



- Object-Oriented Programming
- Classes & Objects
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Working with File Permission

- Files and directories have three level of access:
 - User
 - Group
 - Other
- The three typical permission for files and directories are:
 - Read (r)
 - Write (w)
 - Execute (x)

Working with File Permission

Permissions	First Digit (Leftmost) Always 0	Second Digit User (u)	Third Digit Group (g)	Fourth Digit (Rightmost) Other (o)
Read (r)	0	4	4	4
Write (w)	0	2	2	2
Execute (x)	0	1	1	1

Table 5-2

Octal values for the *mode* parameter of the `chmod()` function

Working with File Permission

- The `chmod()` function is used to change the permission of modes of a file or directory
- The syntax for the `chmod()` function is `chmod($filename, $mode);`
- Where `$filename` is the name of the file to change and `$mode` is an integer specifying the permission for the file

Reading Directories

The following table lists all the PHP functions that read the names of files and directories

Function	Description
<code>chdir(<i>directory</i>)</code>	Changes to the specified directory
<code>chroot(<i>directory</i>)</code>	Changes the root directory of the current process to the specified directory
<code>closedir(<i>handle</i>)</code>	Closes a directory handle
<code>getcwd()</code>	Gets the current working directory
<code>opendir(<i>directory</i>)</code>	Opens a handle to the specified directory
<code>readdir(<i>handle</i>)</code>	Reads a file or directory name from the specified directory handle
<code>rewinddir(<i>handle</i>)</code>	Resets the directory pointer to the beginning of the directory
<code>scandir(<i>directory</i>[, <i>sort</i>])</code>	Returns an indexed array containing the names of files and directories in the specified directory

Table 5-3 PHP directory functions

Reading Directories

- To iterate through the entries in a directory, open a handle to the directory with the `opendir()` function
- Use the `readdir()` function to return the file and directory names from the open directory
- Use the `closedir()` function to close a directory handle

Reading Directories

```
$Dir = "C:\\PHP";  
$DirOpen = opendir($Dir) ;  
while ($CurFile = readdir($DirOpen)) {  
    echo $CurFile . "<br />";  
}  
closedir($DirOpen) ;
```

Reading Directories

```
$Dir = "C:\\PHP";  
$DirOpen = opendir($Dir) ;  
while ($CurFile = readdir($DirOpen)) {  
    echo $CurFile . "<br />";  
}  
closedir($DirOpen) ;
```

File Upload

```
<form action="01_file_upload/proses.php"
        method="post"
        enctype="multipart/form-data">
    <input type="file" name="foto" />
    <button type="submit">Upload</button>
</form>
```

```
<?php
echo "<pre>";
var_dump($_FILES);
```

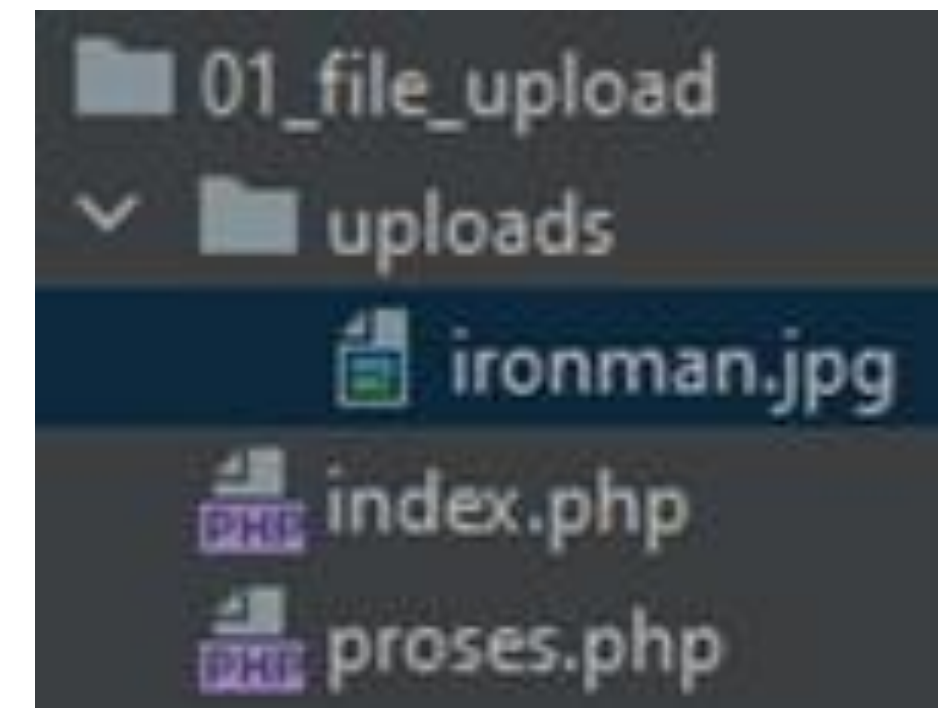
```
array(1) {
    ["foto"]=>
    array(6) {
        ["name"]=>
        string(12) "ironman.webp"
        ["full_path"]=>
        string(12) "ironman.webp"
        ["type"]=>
        string(10) "image/webp"
        ["tmp_name"]=>
        string(24) "C:\xampp\tmp\php4D64.tmp"
        ["error"]=>
        int(0)
        ["size"]=>
        int(90418)
    }
}
```

File Upload

```
<?php

$filename = $_FILES['foto']['name'];
$temp_file = $_FILES['foto']['tmp_name'];

move_uploaded_file($temp_file, "uploads/{$filename}");
```



File Upload Validation

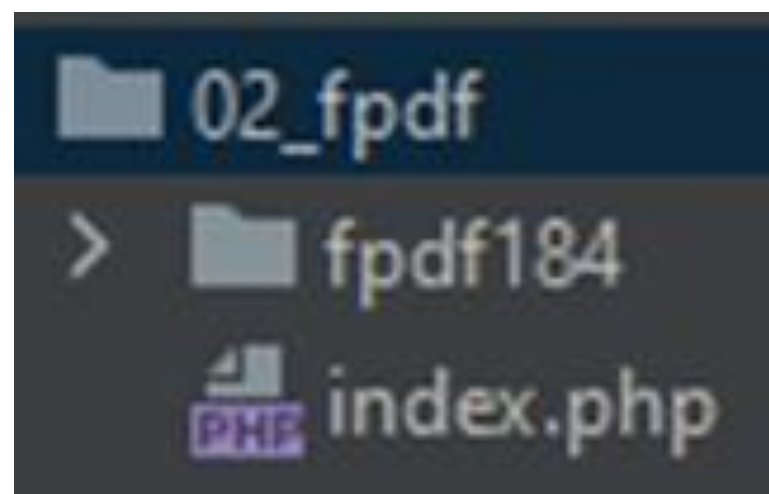
```
<?php
$filename = $_FILES['foto']['name'];
$temp_file = $_FILES['foto']['tmp_name'];

$file_ext = explode(separator: ".", $filename);
$file_ext = end( &array: $file_ext);
$file_ext = strtolower($file_ext);

switch($file_ext){
    case 'jpg':
    case 'jpeg':
    case 'png':
    case 'svg':
    case 'webp':
    case 'bmp':
    case 'gif':
        move_uploaded_file($temp_file, "uploads/{$filename}");
        echo "File berhasil di upload.";
        break;
    default: echo "Anda hanya bisa upload file gambar.";
}
```


Generate PDF with FPDF

Download library FPDF: <http://www.fpdf.org/>



```
<?php
require('fpdf184/fpdf.php');

$pdf = new FPDF();
$pdf->AddPage();
$pdf->SetFont( family: 'Arial', style: 'B', size: 16);
$pdf->Cell( w: 40, h: 10, txt: 'Hello World!');
$pdf->Output();
```


Using Composer

```
alexwawo\03_composer> composer init
```

```
This command will guide you through creating your composer.json config.
```

```
Package name (<vendor>/<name>) [alexwawo/03_composer]:
```

```
Description []: Praktik menggunakan Composer
```

```
Author [alexwawo <alex.wawo@gmail.com>, n to skip]:
```

```
Minimum Stability []: dev
```

```
Package Type (e.g. library, project, metapackage, composer-plugin) []:
```

```
License []:
```

```
Define your dependencies.
```

```
Would you like to define your dependencies (require) interactively [yes]? n
```

```
Would you like to define your dev dependencies (require-dev) interactively [yes]? n
```

```
Add PSR-4 autoload mapping? Maps namespace "Alexwawo\03Composer" to the entered relative path. [s]
```

Simple Hello World with Composer

```
<?php
namespace HelloWorld;

class Greetings {
    public static function sayHelloWorld() {
        return 'Hello World';
    }
}
```

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

use HelloWorld\Greetings;
echo Greetings::sayHelloWorld();
```


Create Excel File with PhpSpreadsheet

```
<?php
require_once __DIR__ . '/vendor/autoload.php';

use PhpOffice\PhpSpreadsheet\Spreadsheet;
use PhpOffice\PhpSpreadsheet\Writer\Xlsx;
use PhpOffice\PhpSpreadsheet\IOFactory;

$spreadsheet = new Spreadsheet();
$sheet = $spreadsheet->getActiveSheet();
$sheet->setCellValue( pCoordinate: 'A1', pValue: 'Hello World!');
$sheet->setCellValue( pCoordinate: 'A2', pValue: 'Hello UMN');
$sheet->setCellValue( pCoordinate: 'A3', pValue: 'Hello Indonesia');
$sheet->setCellValue( pCoordinate: 'B1', pValue: 'John Thor');
$sheet->setCellValue( pCoordinate: 'B2', pValue: 'John Wick');
$sheet->setCellValue( pCoordinate: 'B3', pValue: 'John Doe');

$writer = new Xlsx($spreadsheet);
$writer->save( pFilename: 'hello.xlsx');
```

SUMMARY

After finishing this module, you should be able:

- to understand what is object-oriented programming
- to understand how to implement classes and objects in PHP
- to understand how to implement constructor and destructor in a class
- to understand how to implement polymorphism using access modifier
- to understand how to implement class inheritance
- to understand how to implement interfaces, abstract classes, traits, static properties and methods.
- to understand how to create autoloader
- to understand how to implement namespaces

NEXT WEEK'S OUTLINE

- File Upload
- Read & Write File
- Generate PDF & Excel File

Note: Please install **composer**

<https://getcomposer.org/>



REFERENCES

- <https://www.w3schools.com/PHP/default.asp>
- <https://www.php.net/docs.php>
- Luke Welling & Laura Thomson. 2017. PHP and MySQL Web Development (5th ed)
- Robin Nixon. 2021. Learning PHp, MySQL & JavaScript.

Visi

Menjadi Program Studi Strata Satu Informatika **unggulan** yang menghasilkan lulusan **berwawasan internasional** yang **kompeten** di bidang Ilmu Komputer (*Computer Science*), **berjiwa wirausaha** dan **berbudi pekerti luhur**.



Misi

1. Menyelenggarakan pembelajaran dengan teknologi dan kurikulum terbaik serta didukung tenaga pengajar profesional.
2. Melaksanakan kegiatan penelitian di bidang Informatika untuk memajukan ilmu dan teknologi Informatika.
3. Melaksanakan kegiatan pengabdian kepada masyarakat berbasis ilmu dan teknologi Informatika dalam rangka mengamalkan ilmu dan teknologi Informatika.